



Improved polarization retention in LiNbO₃ single-crystal memory cells with enhanced etching angles

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ABSTRACT

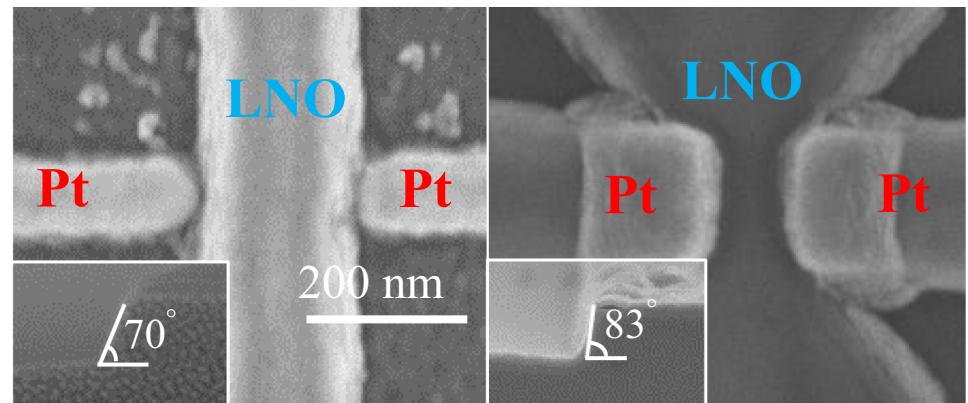
Multifunctional LiNbO₃ material plays an important role in domain wall microelectronics and nonlinear optoelectronics. However, the material is hard and relatively inert, and hence is quite difficultly etched. A new oblique method to etch LiNbO₃ memory cells at the surface of X-cut bulk crystals was proposed in this study. The process includes mask fabrication, oblique etching, and wet corrosion cleaning. The etching angle highly approaches 83° to achieve better polarization retention than others with etching angles of 0° and 70°. Meantime, the measured domain switching hysteresis loops become more symmetrical along the voltage axis than others with the presence of a strong imprint field. The horizontal electric field simulation along the polar Z axis exhibits the enhanced nucleating fields of the domains located at two edges of each memory cell in contact to Pt electrodes under the same applied voltage. The depolarization field was better screened in the memory with a higher etching angle. The new oblique method can improve the performance of the ferroelectric domain wall memory significantly.

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GRAPHICAL ABSTRACT



Introduction

Ferroelectric materials with spontaneous polarizations have piezoelectric, electro-optic, and pyroelectric properties that are widely used in nonvolatile memories, sensors, and electro-optic modulators [1–3]. For memory, a long retention time, a low switching voltage, and a large read current are the core parameters. Many works have been done to improve the retention: Kang et al. [4] used Ir/IrO₂ electrodes in Pb(Zr_xTi_{1-x})O₃ (PZT) thin films; Liu et al. [5] used high-field charge injection in highly strained BiFeO₃ thin films; and Xu et al. [6] employed GdScO₃ (110) substrates to reduce the lattice misfit in epitaxial BiFeO₃ thin films. However, these materials have some shortcomings, for example, PZT can pollute the environment, and the BiFeO₃ deposition at high temperature is not compatible with complementary metal oxide semiconductor (CMOS) technology for the mass production.

In contrast, a multifunctional lithium niobate (LNO) material has wide applications not only in optical fields [7, 8], but also in the field of nonvolatile memories. Conducting ferroelectric domain walls (DWs) in insulating LNO have been reported recently [9], enabling the integration of either a nonvolatile DW memory or a field-effect transistor for the development of nanodomain wall microelectronics [10–13]. Most importantly, the success in the fabrication of CMOS-compatible LNO single-crystal thin

films bonded to SiO₂/Si substrates makes the mass production possible [14].

However, LNO thin-film capacitors using top and bottom electrodes have poor polarization retention once operated in a nanosecond time scale, where the switched polarizations would return back into their initial orientations during retention time due to the presence of a strong imprint field [15]. With the fabrication of the mesa-like cell to contact two side Pt electrodes at the film surface, the nonvolatile binary information can be written through the formation of parallel/antiparallel domains between the etched domain at the top and the referencing domain at the bottom under application of in-plane positive/negative electric fields (E). The written information can be later read out via “on”/“off” currents flowing through erasable conducting DWs between them at a small read voltage [15]. However, the etching of the hard and relatively inert LNO mesas has always been an arduous task [16]. In particular, a small etching angle of a mesa could arouse poor polarization retention and a strong depolarization field in the device. The depolarization field could result in an asymmetric domain switching hysteresis loop and a high coercive electric field (E_c) to deteriorate the memory performance [17].

There are many micromachining techniques to fabricate the LNO mesas, such as mechanical tools, laser ablation, wet etching, ion beam etching, and Ar gas-based plasma sputtering [18–20], among which wet and dry etching technologies are the most used.

These etching techniques have been improved in combination with other auxiliary methods, including proton exchange [21], ion implantation [22, 23], and so on. However, wet etching after proton exchange that damages the LNO ferroelectricity is anisotropic and time-consuming and cannot satisfy the requirement of the good information storing ability during the mass production.

In comparison, dry etching has many advantages over wet etching because of its high selectivity and controllability. These techniques include direct laser writing [24], focused ion beam milling [25], and reactive ion etching (RIE) [26]. However, many of them use fluorine-containing gases (like CF_4 , SF_6 , and CHF_3) which can form the LiF residual. It is well known that the inert LiF can hinder the further etching of LNO, resulting in a small etching angle of the cell.

In this paper, a new oblique etching technology was proposed to fabricate LNO memory cells, making the etching angles (θ) as high as 83° . The key of the success lies in sample inclination during etching that can gather more ions to bombard the sidewalls of the mesas. Meantime, the by-products can be removed in time to permit steep etching of the LNO effectively. Finally, the electrical characterization confirms the improved polarization retention and reduced E_c .

Experimental method

Oblique etching method

The commercial congruent X-cut LNO single crystals were bought from Sciences Opto Electronic Materials and Equipment Dept, Shanghai Institute of Optics Fine Mechanics, and were grown with Czochralski technology using high-purity (99.99%) Li_2CO_3 , MgO , and Nb_2O_5 melted powders at 1250°C . The above crystal containing 48.5 mol% Li_2O and 5 mol% MgO was electrically polarized in a single domain pattern at $1180\text{--}1200^\circ\text{C}$ at a current density of $8\text{--}10\text{ mA cm}^{-2}$ for 30 min and then cut into the sizes of $10 \times 10 \times 0.5\text{ mm}^3$ with both sides polished. The oblique etching process of LNO can be divided into three steps: mask fabrication, oblique etching, and wet corrosion cleaning. Figure 1a–f briefly describes the etching process.

Mask fabrication

The electroplated Ni was chosen as a hard mask to resist fluorine-based plasma etching and wet corrosion cleaning of LNO under protecting area. 20-nm-thick Cr and 40-nm-thick Au films were firstly grown as seed layers using magnetron sputtering (PVD-75, Kurt J. Lesker) on the surfaces of 5 mol% MgO -doped X-cut LNO single crystals at room temperature. A 200-nm-thick poly (methyl methacrylate) photoresist layer was then spin-coated on the surface, and the sizes of LNO mesas were then defined and patterned using electron-beam lithography (EBL JEOL 6300FS), as shown in Fig. 1a. After that, the 100-nm-thick Ni layer was electroplated on the exposed seed layer without the photoresist, as shown in Fig. 1b. Finally, the photoresist was removed completely using an acetone bath, as shown in Fig. 1c.

Oblique etching

A reactive ion etching system (RIE-10NR, Samco) was used to etch the seed layer and LNO under optimized conditions [27, 28]. Firstly, Ar and SF_6 /Ar gases were employed to etch Au and Cr in sequence. When the seed layer was etched completely, the LNO was etched using Ar and SF_6 mixed gases with an etching rate of 8 nm min^{-1} in three steps: (1) etching for 6 min in a normal angle (Fig. 1d); (2) etching for 3 min in an inclined angle of -15° (Fig. 1e); and (3) etching for 3 min in an inclined angle of 15° (Fig. 1f).

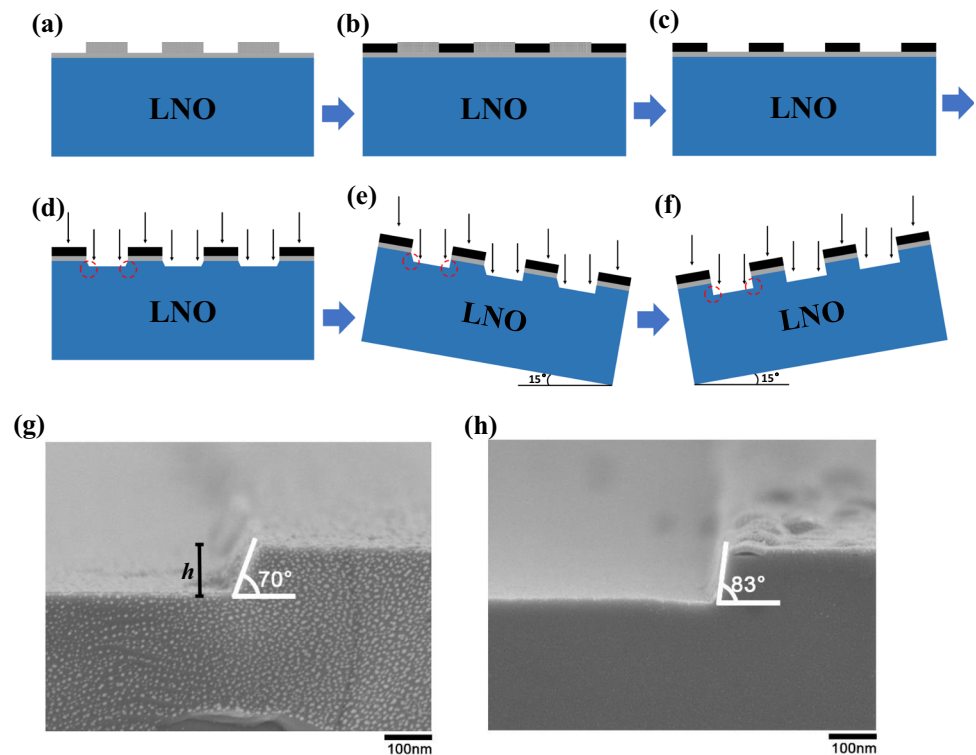
Wet corrosion cleaning

After etching, the sample was immersed into the NH_4OH , H_2O_2 and H_2O (1:1:1) bath at 85°C to remove the damaged layers and redepositions [29]. Figure 1g, h show the SEM images of the sidewalls using normal etching and oblique etching methods, respectively. It is clear that the latter increases the sidewall angle from 70° to 83° .

Memory cell fabrication

After the mesa fabrication, Ni mask and Cr/Au seed layers were removed using corrosion solutions. Subsequently, 30-nm-thick Pt top electrode layer was grown at 300°C using the magnetron sputtering technique. Then, the Pt layer was etched into left and right electrodes (L and R) with the width of w to

Figure 1 High-angle etching process of LNO cells: **a** seed layer deposition and patterning using EBL; **b** Ni electroplating; **c** photoresist removal using the acetone bath; **d, e, f** three-step etching of the seed layer and LNO in different inclined angles; **g, h** cross-section SEM images of two LNO cells without/with substrate inclination during RIE etching, respectively.



contact the LNO mesas by repeating EBL patterning and RIE etching. To mitigate etching damage to L and R contacts to LNO, additional 150-nm-thick Ni electrodes were electroplated as mask above Pt. All memory cells defined by dimensions of $w \times l \times h$ were imaged using scanning electron microscopy images (SEM; Sigma HD, Zeiss).

Testing process

Each double current-voltage (I - V) curve was measured using an Agilent B1500A semiconductor analyzer operated in voltage-sweep mode with sweeping time of 1s. The domain patterns were characterized after voltage poling via in-plane piezoresponse force microscopy (PFM; Icon, Bruker) using a contact PtIr-coated silicon tip with a radius of ~ 20 nm and AC amplitudes of 0.5–2 V at 230 kHz.

Electric field simulation

Equation $E_i = -\partial\phi/\partial x_i$ was used to calculate the electric field, and ϕ was the electrical potential. For obtaining ϕ , the Poisson equation $\nabla^2\phi = \rho$ was solved by using the specified electric potentials at L and R. On the surface beyond the electrodes, an open-circuit boundary condition was appended. A

nonlinear multifield coupling finite-element method was employed for solving the equations above.

Results and discussion

Influence of etching angles

Figure 2a–c show the planar SEM images of three memory cells in sizes of $w \times l \times h = 200 \times 80 \times 0$ nm³, $100 \times 200 \times 70$ nm³, $200 \times 150 \times 100$ nm³ and etching angles of 0°, 70°, and 83°, respectively. Figure 2d–f shows their corresponding I - V curves, where the arrows indicate voltage sweeping directions. When a write (poling) voltage slightly higher than a forward coercive voltage (V_{c+}) is applied to L with R grounded, the domains between L and R will be switched oppositely against the unswitched referencing domain at the bottom, and the DWs will form between them. This accompanies the abrupt off-to-on current jump in the first I - V sweep. When the write (poling) voltage is lower than a backward coercive voltage (V_{c-}), the switched domain will return to its original state and the DWs will disappear. In the next cycles, this process is repeated. The current is diode-like, and V_{c+} is 12.5 V, 13.5 V and 8 V for the three samples, as inferred from their first

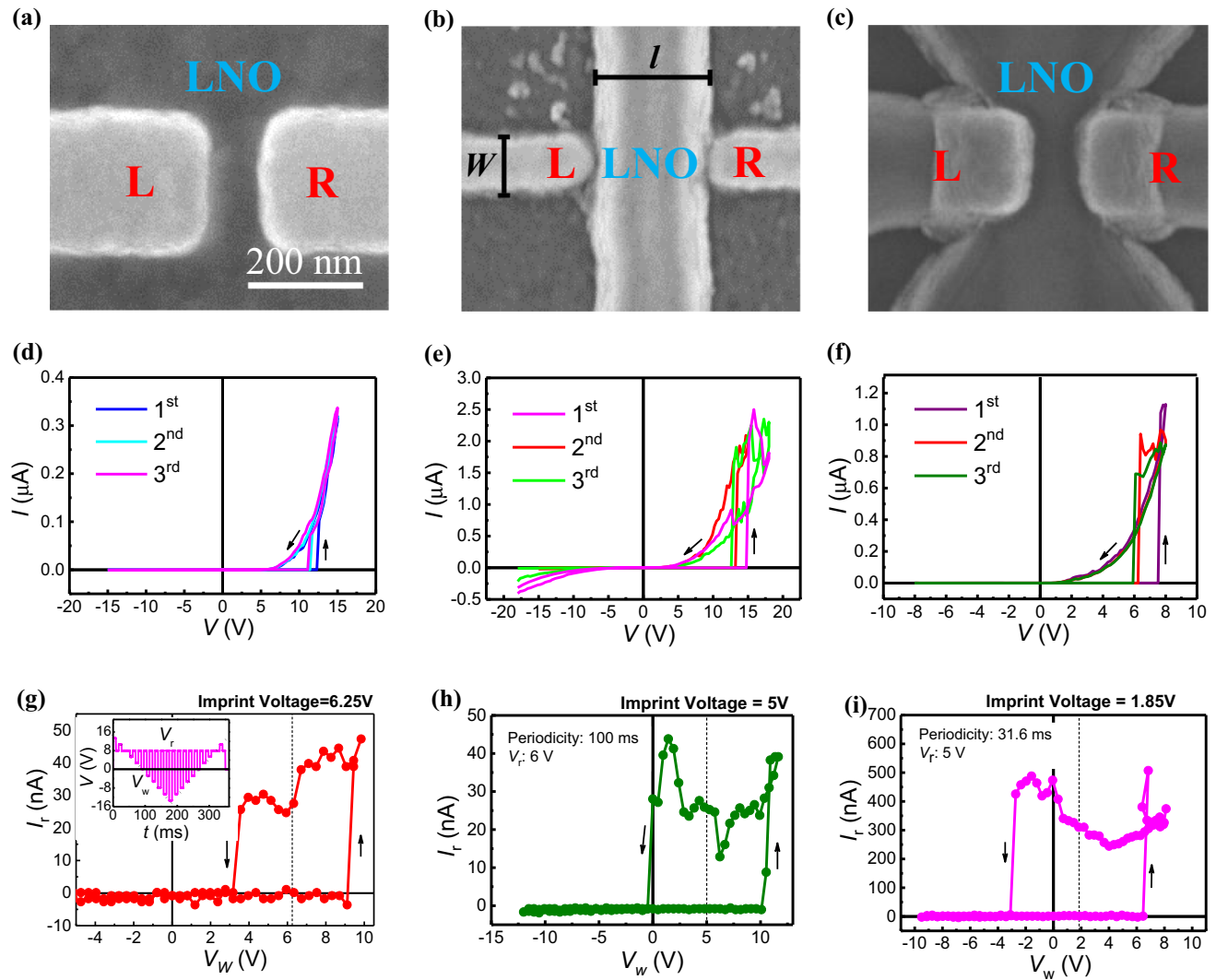


Figure 2 **a, b, c** Planar SEM images of the mesas in sizes of $w \times l \times h = 200 \times 80 \times 0 \text{ nm}^3$, $100 \times 200 \times 70 \text{ nm}^3$ and $200 \times 150 \times 100 \text{ nm}^3$ with the etching angles of 0° , 70° and 83° , respectively. **d, e, f** Corresponding I - V curves after different

sweeping cycles. **g, h, i** Corresponding $I_r - V_w$ hysteresis loops of the cells with the interweaved write-read voltage ($V_w - V_r$) pulses shown in the inset. Thin arrows show voltage sweeping directions.

I - V sweeps, respectively. V_{c+} reduces a little and gradually levels off after next two cycles.

Figure 2g–i show domain switching hysteresis loops for the three devices with the etching angles of 0° , 70° and 83° , respectively. The loops were characterized with the interposed write and read pulses shown in the inset of Fig. 2g. In these tests, the write pulses in voltages of V_w were programmed in a triangular waveform shape with changing amplitudes of $+10\text{V} \rightarrow -5\text{V} \rightarrow +10\text{V}$ in Fig. 2g, $+12\text{V} \rightarrow -12\text{V} \rightarrow +12\text{V}$ in Fig. 2h, and $+8\text{V} \rightarrow -10\text{V} \rightarrow +8\text{V}$ in Fig. 2i in periodicities of 350 ms, 100 ms, and 31.6 ms, respectively. A read pulse with constant voltage of V_r was inserted between two adjacent

write pulses to read wall current (I_r) against V_w . Figure 2g shows the $I_r - V_w$ hysteresis loop as $\theta = 0^\circ$ from which we derived $V_{c+} = 9.4 \text{ V}$ and $V_{c-} = 3.16 \text{ V}$. The loop is asymmetric due to the presence of a strong imprint voltage of 6.25 V, and $V_{c+} \cdot V_{c-} > 0$ implies the occurrence of domain back-switching (poor retention) as the write voltage is reduced to 0 V. However, the imprint voltage reduces to 5.0 V as $\theta = 70^\circ$ in Fig. 2h and 1.85 V as $\theta = 83^\circ$ in Fig. 2i with $V_{c+} \cdot V_{c-} < 0$, indicating better polarization retention with increasing etching angle.

From the first I - V sweeps, we derived the V_{c+} dependence on l , which could be fitted by a solid line, as shown in Fig. 3. From the slopes of these linear fits,

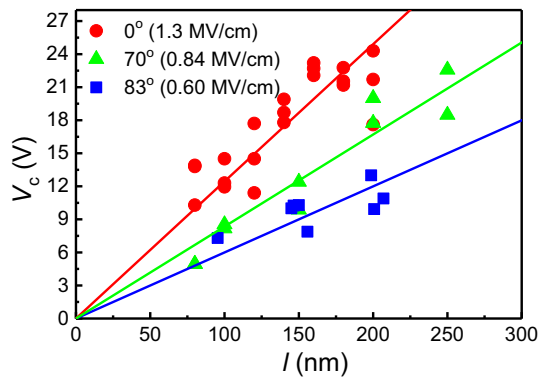


Figure 3 Length dependence of coercive voltage inferred from the first I - V sweeps for LNO mesas in different etching angles. All plots can be fitted by the linear lines with the values of the slopes shown in parentheses.

E_c was estimated to 1.3 MV cm^{-1} , 0.84 MV cm^{-1} , and 0.60 MV cm^{-1} as $\theta = 0^\circ$, 70° , and 83° , respectively. It can be concluded that E_c decreases with the inclined etching angle, enabling an energy-efficient domain wall memory in a fast operation speed and a low write voltage.

Domain switching behavior

The switched domains can be observed from the PFM images. Figure 4a shows in-plane PFM phase and amplitude images of the inversed domains across the cell in sizes of $w \times l \times h = 100 \times 200 \times 70 \text{ nm}^3$ and $\theta = 70^\circ$ after poling at $\pm 14 \text{ V}$. The region in white contrast from the phase image (upper-left panel in Fig. 4a) at $+14 \text{ V}$ indicates the 180° domain reversal (the red and blue arrows show polarization orientations) between L and R electrodes with two thick black lines of the DW regions shown in the amplitude image (upper-right panel in Fig. 4a), accompanying the abrupt off-to-on current jump in the I - V sweep (Fig. 2e). After poling voltage at -14 V , the white contrast region from the phase image and the black-line DW regions from the amplitude image both disappear (lower panels in Fig. 4a), predicting the domain back-switching into its previous orientation. The similar reversible domain switching behavior between R and L was also observed for another cell with $\theta = 83^\circ$ after poling at $\pm 20 \text{ V}/-12 \text{ V}$, as shown in Fig. 4b. However, the domain sideways motion beyond the R and L area is obvious in upper panels of Fig. 4b, and some segments remain after the erasure of the domain at an opposite voltage of -12 V (lower panels in Fig. 4b). For the sample without etching

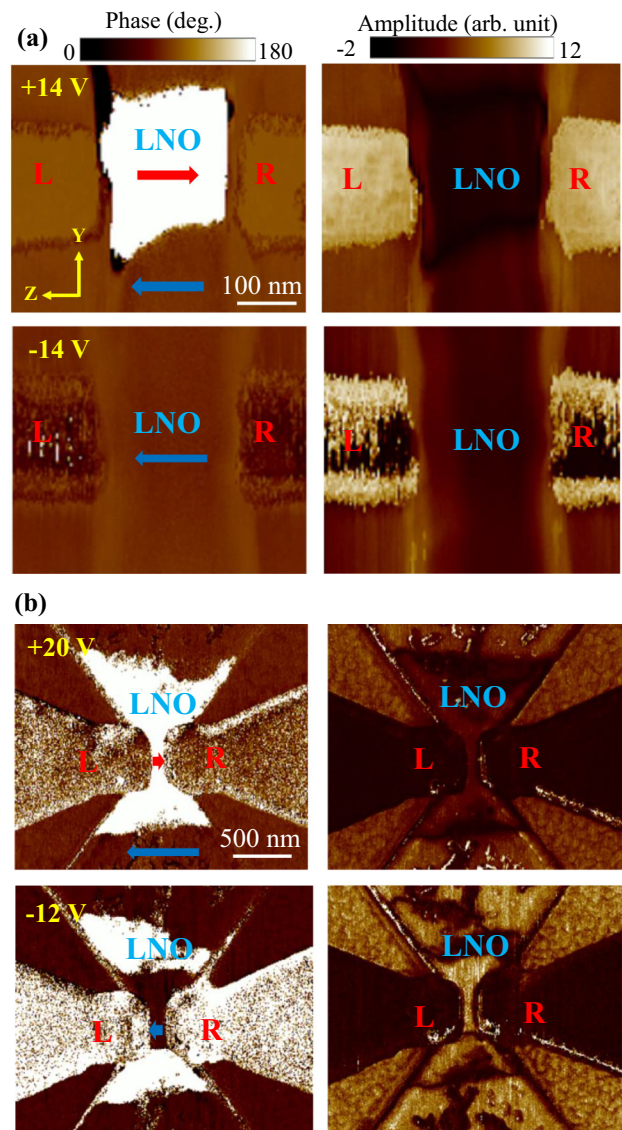


Figure 4 **a** In-plane PFM phase (left panels) and amplitude (right panels) images of switched domains within a mesa in sizes of $w \times l \times h = 100 \times 200 \times 70 \text{ nm}^3$ and an etching angle of 70° after poling voltages of $\pm 14 \text{ V}$. The red/blue arrows show polarization orientations after/before domain switching. **b** In-plane PFM phase (left panels) and amplitude (right panels) images of switched domains within a mesa in sizes of $w \times l \times h = 374 \times 350 \times 160 \text{ nm}^3$ and an etching angle of 83° after poling voltages of $+20 \text{ V}/-12 \text{ V}$.

($\theta = 0^\circ$), it was failed to observe the switched domain from PFM images due to the poor polarization retention.

Figure 5a shows the pulse sequence to characterize domain switching time. After application of a pre-setting pulse in voltage/width of $10 \text{ V}/1 \text{ ms}$, a write pulse in voltage/width of V_{sw}/t is applied to switch

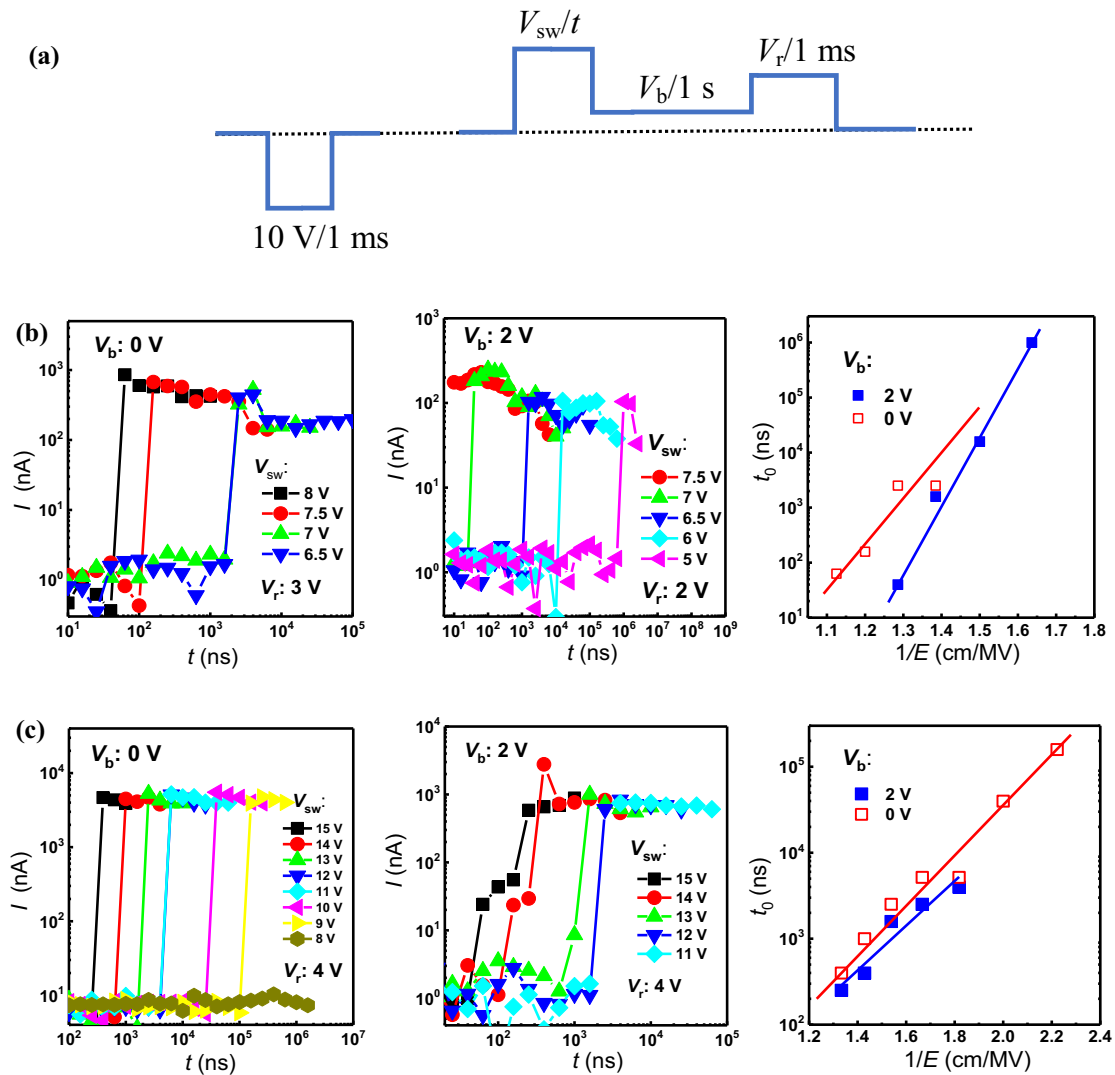


Figure 5 a Sketched pulse sequence for testing of domain switching time. **b** Write time dependence of wall current for a memory cell in sizes of $w \times l \times h = 400 \times 90 \times 145$ nm³ and an etching angle of 70° after different V_{sw} and V_b in left and middle panels. The rightmost panels in **b** and **c** show their inversed electric field dependences of domain switching time in semilogarithmic plots fitted by the solid lines. **c** Write time dependence of wall current for a

memory cell in sizes of $w \times l \times h = 220 \times 200 \times 160$ nm³ and an etching angle of 83° after different V_{sw} and V_b in left and middle panels. The rightmost panels in **b** and **c** show their inversed electric field dependences of domain switching time in semilogarithmic plots fitted by the solid lines. The domain switching time t_0 shortens with increasing V_{sw} . The rightmost panels show their t_0 – $1/E$ dependences that can be described using the Merz law of $t_0 \sim \exp(E_a/E)$ [30, 31], as confirmed by the

solid line fits of the data in semilogarithmic plots, where E_a is the activation field. The domain switching speed when $V_b = 0$ V is much slower than that when $V_b = 2$ V in Fig. 5b ($\theta = 70^\circ$), especially at high V_{sw} . Nevertheless, the slower write operation when $V_b = 0$ V can be improved in Fig. 5c when $\theta = 83^\circ$ where the switched domain has better polarization retention (Fig. 2i).

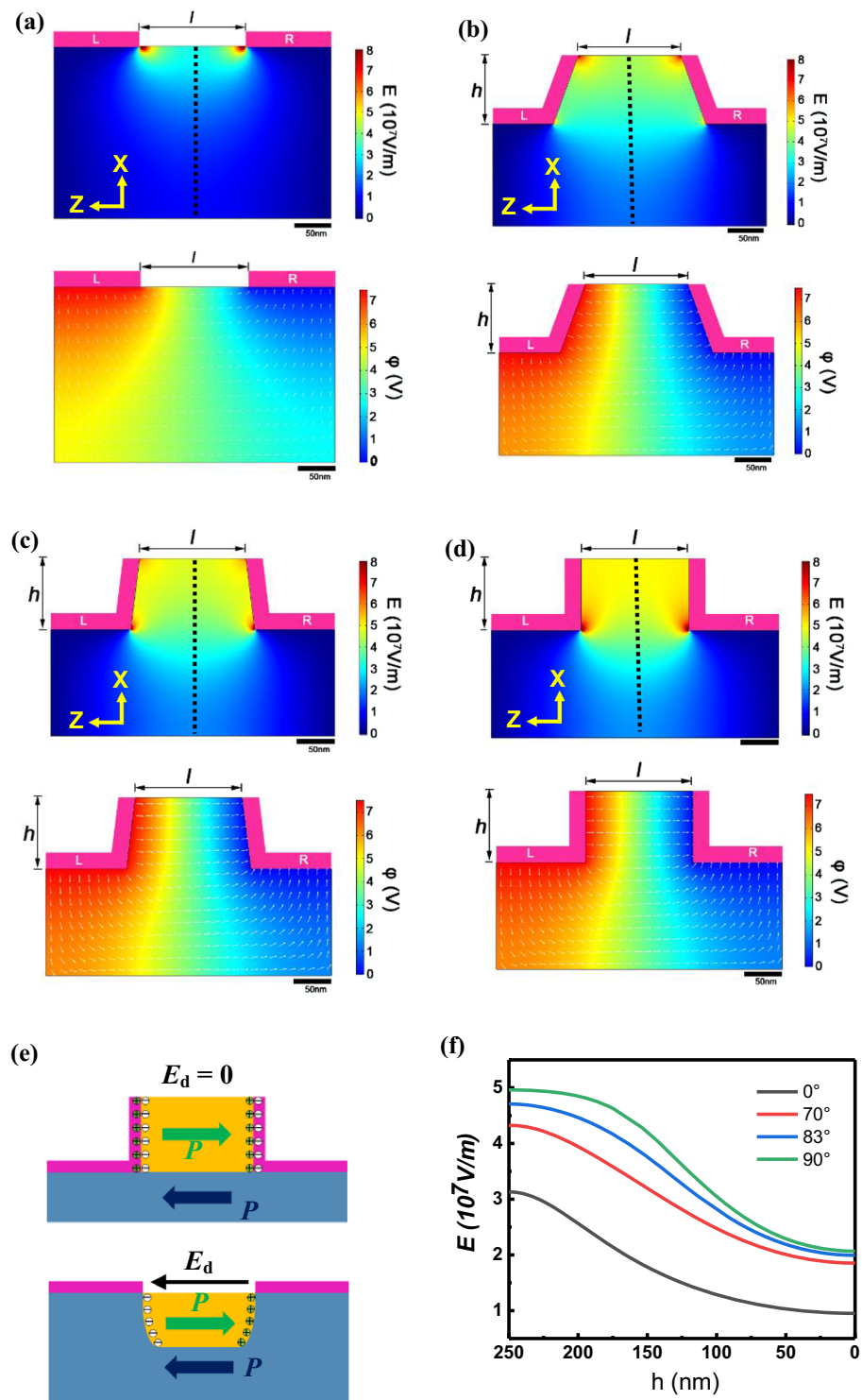
Electric field and potential distributions

For the understanding of the etching angle effect on the polarization retention and coercive voltage, the

horizontal electric field and potential distributions were simulated for the cells with the sizes of $l = 150$ nm, $h = 100$ nm, and $\theta = 0^\circ, 70^\circ, 83^\circ$ and 90° at 7.5 V, as shown in Fig. 6a–d, respectively. Generally, the domain switching initiates from the reversed

domain nucleation at positions near the LNO contacting to L and R electrodes, where the nucleating field is maximum. As $\theta = 0^\circ$ (Fig. 6a), the nucleating energy and E_c are highest for the domain growth along the film surface. With increasing etching angles

Figure 6 a–d Horizontal electric field and potential distributions for the cells with the sizes of $l = 150$ nm and $h = 100$ nm at 7.5 V with etching angles of $0^\circ, 70^\circ, 83^\circ$ and 90° , respectively. e Schematic diagrams of charge distributions and depolarization fields in etched and unetched samples. f Depth dependences of horizontal electric field along the dotted lines in (a–d).



from Fig. 6b–d, the maximum fields appear at positions near the bottom-left and bottom-right corners where the domains have the lowest nucleating energy and E_c . Meantime, the depolarization field (E_d) arising from the domain boundary charge is better screened by L and R electrodes as θ approaches 90° as shown in Fig. 6e (higher panel). Therefore, the imprint field in Fig. 2c is smallest to achieve better polarization retention for all switched domains. Otherwise, the switched domain boundary charge becomes uncompensated as $\theta = 0$ in Fig. 6e (lower panel), where the depolarization field is strongest. This unscreened E_d destabilizes the switched domain within the L and R gap in Fig. 2a where the domain shows poor polarization retention [32]. Figure 6f shows etching depth dependences of horizontal electric field along the dotted lines in Fig. 6a–d. It can be seen that horizontal electric field increases as the etching angle enhances in Fig. 6f, indicating the increase of the effective nucleating field projected along the polar Z axis to reduce E_c in Fig. 3.

Conclusion

In this study, a new oblique method to etch the LNO memory cell was proposed with the etched sidewall angle as high as 83° . In comparison to the angles of 0° and 70° , the enhanced etching angle can improve polarization retention and reduce the coercive field. Subsequent I_r – V_w hysteresis loops and PFM images show good retention of the switched domains as $\theta \geq 70^\circ$. Finally, the finite-element simulations show the maximum nucleating field at the film surface as $\theta = 0^\circ$ where the nucleating energy and E_d are highest, in comparison to the lowest nucleating energy and E_d at the positions near the bottom-left and bottom-right corners across the cell as $\theta = 90^\circ$. These simulations are in agreement with our experimental characterization. This oblique etching technique to enhance sidewall angle is not only beneficial for developing of next generation domain wall nanodevices in fast operation speeds and low-power consumptions, but also helpful for the manufacture of other electro-optic devices based on the LNO material.

Acknowledgements

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